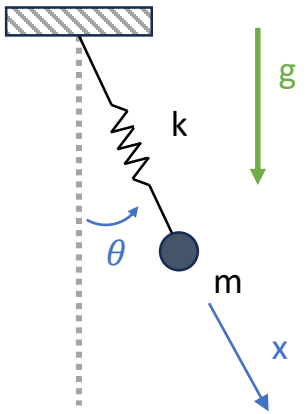
Structures and Materials

Home Task | Dr.-Ing. Andreas Hermanutz

Elementary Concepts of Elasticity

3. Elastic Pendulum



Task: Derive the equations of motion for the elastic pendulum.

Kinetic Energy:

$$T = \frac{1}{2} m \vec{v}^2$$

$$T = \frac{1}{2} m \left(\dot{x}^2 + \left(l\dot{\theta} \right)^2 \right) \qquad l(t) = l_0 + x$$

$$T = \frac{1}{2} m \left(\dot{x}^2 + (l_0 + x)^2 \dot{\theta}^2 \right)$$

Potential Energy:

$$\Pi = mgh + \frac{1}{2} kx^2$$

$$\Pi = -mg(l_0 + x)cos(\theta) + \frac{1}{2} kx^2$$

3. Elastic Pendulum

Lagrange:

$$\mathbf{L} = \mathbf{T} - \mathbf{\Pi}$$

$$\mathbf{L} = \frac{1}{2}m\left(\dot{x}^2 + (l_0 + x)^2\dot{\theta}^2\right) + mg(l_0 + x)cos(\theta) - \frac{1}{2}kx^2$$

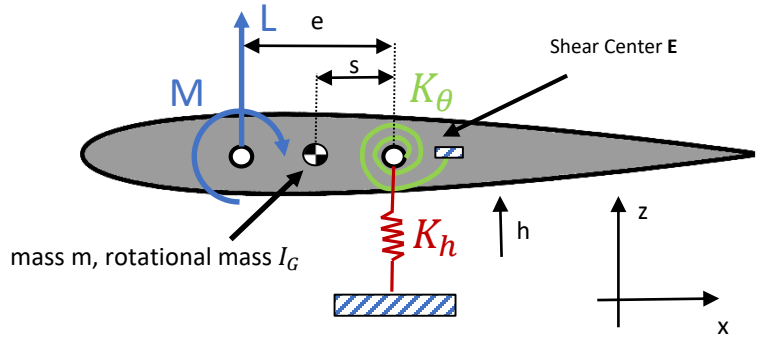
$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = Q$$

x: $$m\ddot{x} - \left(m(l_0 + x)\dot{\theta}^2 + m\,g\,cos(\theta) - kx\right) = 0$$

θ: $$2m(l_0 + x)\dot{x}\dot{\theta} + m\,(l_0 + x)^2\ddot{\theta}^2 + m\,g(l_0 + x)\,sin(\theta) = 0$$

$$\begin{Bmatrix}\ddot{x}\\ \ddot{\theta}\end{Bmatrix} = \begin{bmatrix}(l_0 + x)\dot{\theta}^2 - \frac{K}{m}x + g\,cos(\theta)\\ \frac{-2}{(l_0+x)}\dot{x}\dot{\theta} - \frac{g}{(l_0+x)}sin(\theta)\end{bmatrix}$$

4. Elastic Wing Segment



Task: Find the equations of motion for the elastic wing segment assuming small displacements.

Potential Energy:

$$\Pi = \frac{1}{2}K_h h^2 + \frac{1}{2}K_\theta \theta^2$$

Kinetic Energy:

$$T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\dot{\theta}^2$$

$$\vec{X}_G = \begin{bmatrix} s \cdot \cos(\theta) \\ s \cdot \sin(\theta) + X_E \end{bmatrix}$$

$$\vec{V}_G = \begin{bmatrix} -s \cdot \sin(\theta)\dot{\theta} \\ s \cdot \cos(\theta)\dot{\theta} + \dot{h} \end{bmatrix}$$

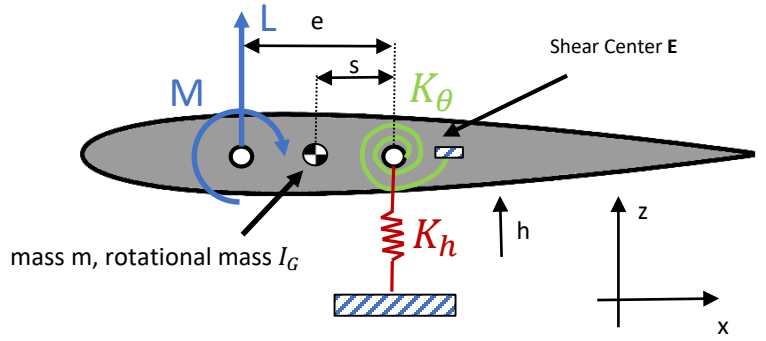
For small θ:

$$\sin(\theta) \approx 0 \qquad \cos(\theta) \approx 1$$

$$\vec{V}_G = \begin{bmatrix} 0 \\ s \cdot \dot{\theta} + \dot{h} \end{bmatrix}$$

$$v_G^2 = \left(s \cdot \dot{\theta} + \dot{h}\right)^2$$

5. Elastic Wing Segment



Task: Find the equations of motion for the elastic wing segment assuming small displacements.

$$T = \frac{1}{2}m\left(s\cdot\dot{\theta}+\dot{h}\right)^2 + \frac{1}{2}I_G\dot{\theta}^2 = \frac{1}{2}m\dot{h}^2 + s\cdot m\cdot\dot{h}\cdot\dot{\theta} + \frac{1}{2}ms^2\dot{\theta}^2 + \frac{1}{2}I_G\dot{\theta}^2 = \frac{1}{2}m\dot{h}^2 + s\cdot m\cdot\dot{h}\cdot\dot{\theta} + \frac{1}{2}\underbrace{\left(I_G+ms^2\right)}_{I_E}\dot{\theta}^2$$

External Work: $W_e = M\cdot\theta + L\cdot h + L\cdot e\cdot\theta$

h: $m\left(\ddot{h}+s\ddot{\theta}\right) + K_h\cdot h = L$

θ: $I_E\ddot{\theta} + m\cdot s\cdot\ddot{h} + K_\theta\cdot\theta = M + e\cdot L$

$$\begin{bmatrix} m & m\cdot s \\ m\cdot s & I_E \end{bmatrix}\begin{Bmatrix}\ddot{h}\\ \ddot{\theta}\end{Bmatrix} + \begin{bmatrix} K_h & 0 \\ 0 & K_\theta \end{bmatrix}\begin{Bmatrix} h \\ \theta\end{Bmatrix} = \begin{Bmatrix} L \\ M + e\cdot L\end{Bmatrix}$$